

PLASTIC-CRAFT

Nylon

Polyamide (PA) — Type 6 & Type 6/6 Grades

845-358-3010
ecom@plastic-craft.com
www.plastic-craft.com

Versatile semi-crystalline engineering thermoplastic with outstanding wear resistance, high strength-to-weight ratio, and self-lubricating properties. Frequently used as a metal replacement for bearings, gears, and structural components in demanding mechanical applications.

STOCK AVAILABILITY

Plastic-Craft maintains extensive inventory across a wide range of standard and specialty sizes in sheets, rods, tubes, and custom profiles. Most orders ship from stock — contact us for immediate availability.

KEY HIGHLIGHTS

- Excellent wear & abrasion resistance
- High strength-to-weight ratio
- Self-lubricating surface
- Good chemical resistance
- FDA compliant grades available
- ⚠ Absorbs moisture — store sealed; allow for dimensional growth.

COMMON APPLICATIONS

- Gears & sprockets
- Bearings & bushings
- Rollers & wheels
- Wear pads & guides
- Electrical insulators
- Food processing equipment
- Structural components

GRADES AVAILABLE

- Type 6 (extruded)
- Type 6/6 (extruded)
- Cast Type 6
- MoS₂-filled (lubricated)
- Oil-filled
- Glass-filled
- Colors (natural/black)

MATERIAL PROPERTIES — TYPICAL VALUES (DRY, AS-MOLDED)

PROPERTY	ASTM	TYPICAL VALUE	UNITS
PHYSICAL			
Specific Gravity	D792	1.14	—
Water Absorption, 24h	D570	1.2	%
Water Absorption, Saturation	D570	7.0	%
MECHANICAL			
Tensile Strength, Yield	D638	12,000	psi (83 MPa)
Tensile Modulus	D638	420,000	psi (2,900 MPa)
Elongation at Break	D638	60	%
Flexural Strength	D790	15,000	psi (103 MPa)
Flexural Modulus	D790	410,000	psi (2,830 MPa)
Compressive Strength	D695	12,500	psi (86 MPa)
Impact, Izod Notched	D256	0.6 – 0.9	ft-lb/in (32–48 J/m)
Rockwell Hardness	D785	R121	Scale R
FRICTION / WEAR			
Coefficient of Friction (Dynamic)	D3702	0.25	vs Steel
Coefficient of Friction (Static)	D3702	0.35	vs Steel
Wear Factor K	D3702	1.5 × 10 ⁻¹⁰	in ³ ·min/ft·lb·hr
THERMAL			
Heat Deflection @ 264 psi	D648	194	°F (90°C)
Heat Deflection @ 66 psi	D648	455	°F (235°C)
Melting Point	—	500	°F (260°C)
CTE (Linear)	D696	4.0 × 10 ⁻⁵	in/in/°F
Thermal Conductivity	C177	1.7	BTU·in/hr·ft ² ·°F
Max Continuous Service	—	220	°F (104°C)
ELECTRICAL			
Volume Resistivity	D257	> 10 ¹⁵	ohm·cm
Surface Resistivity	D257	> 10 ¹³	ohm/sq
Dielectric Strength	D149	400	V/mil (16 kV/mm)
Dielectric Constant, 60Hz	D150	4.0	—
FLAMMABILITY			
UL Flammability Rating	UL 94	V-2	Self-extinguishing
Horizontal Burn Rate	D635	SE	Self-extinguishing
DURABILITY			
Abrasion Resistance (Taber)	D1044	6 – 12	mg/1000 cycles
Weathering / UV	G154	Fair	UV stabilized grades avail.

PROCESSING / FABRICATION NOTES: Excellent machinability with standard tooling | Absorbs moisture — store in sealed packaging | Pre-dry 170°F / 4–6 hr before molding | Allow for dimensional growth in humid environments

CHEMICAL RESISTANCE: Resistant to most hydrocarbons, ketones, esters, alkalis & dilute acids at room temperature. **AVOID:** strong mineral acids (H₂SO₄, HCl, HNO₃), phenols, and formic acid.

Typical values (dry, as-molded); not to be used as specifications. Properties decrease with moisture absorption. Contact Plastic-Craft for grade-specific data.